VALUATIVE CRITERION FOR SEPARATEDNESS OVER A LOCALLY NOETHERIAN BASE, GENERIC-POINT VARIANT

THE STACKS PROJECT AUTHORS

Editorial note. Arrow construction refers to the standard material of Schemes, Section schemes-section-points; not an omission.

Allowed standard prerequisites (editorial).

- Krull-Akizuki DVR existence. Algebra, Lemma algebra-lemma-exists-dvr

Source reference. Original equation 0206: sources/stacks-chow/tags.txt, source lines 2623–2623. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01J5: sources/stacks-chow/tags.txt, source lines 2002–2002. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KP: sources/stacks-chow/tags.txt, source lines 2058–2058. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01OY: sources/stacks-chow/tags.txt, source lines 2211–2211. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01L0: sources/stacks-chow/tags.txt, source lines 2069–2069. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PH: sources/stacks-chow/tags.txt, source lines 931–931. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $f : X \to S$ be a morphism of schemes. Assume f is locally of finite type. The following are equivalent:*

1. *The morphism f is separated.*
2. *For any diagram (0206) there is at most one dotted arrow.*
3. *For all diagrams (0206) with A a discrete valuation ring there is at most one dotted arrow.*
4. *For any irreducible component X_0 of X with generic point $\eta \in X_0$, for any discrete valuation ring $A \subset K = \kappa(\eta)$ with fraction field K and any diagram (0206) such that the morphism $\mathrm{Spec}(K) \to X$ is the canonical one (see Schemes, Section 01J5) there is at most one dotted arrow.*

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Proof. Clearly (1) implies (2), (2) implies (3), and (3) implies (4). It remains to show (4) implies (1). Assume (4). We begin by reducing to S affine. Being separated is a local on the base (see Schemes, Lemma 01KP). Hence, if we can show that whenever $X \to S$ has (4) that the restriction $X_\alpha \to S_\alpha$ has (4) where $S_\alpha \subset S$ is an (affine) open subset and $X_\alpha := f^{-1}(S_\alpha)$, then we will be done. The generic points of the irreducible components of X_α will be the generic points of irreducible components of X, since X_α is open in X. Therefore, any two distinct dotted arrows in the diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X_\alpha \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & S_\alpha \end{array} \tag{0.1.1}$$

would then give two distinct arrows in diagram (0206) via the maps $X_\alpha \to X$ and $S_\alpha \to S$, which is a contradiction. Thus we have reduced to the case S is affine. We remark that in the course of this reduction, we prove that if $X \to S$ has (4) then the restriction $U \to V$ has (4) for opens $U \subset X$ and $V \subset S$ with $f(U) \subset V$.

We next wish to reduce to the case $X \to S$ is finite type. Assume that we know (4) implies (1) when X is finite type. Since S is Noetherian and X is locally of finite type over S we see X is locally Noetherian as well (see Morphisms, Lemma 01T6). Thus, $X \to S$ is quasi-separated (see Properties, Lemma 01OY), and therefore we may apply the valuative criterion to check whether X is separated (see Schemes, Lemma 01L0). Let $X = \bigcup_\alpha X_\alpha$ be an affine open cover of X. Given any two dotted arrows, in a diagram (0206), the image of the closed points of $\mathrm{Spec}\, A$ will fall in two sets X_α and X_β. Since $X_\alpha \cup X_\beta$ is open, for topological reasons it must contain the image of $\mathrm{Spec}(A)$ under both maps. Therefore, the two dotted arrows factor through $X_\alpha \cup X_\beta \to X$, which is a scheme of finite type over S. Since $X_\alpha \cup X_\beta$ is an open subset of X, by our previous remark, $X_\alpha \cup X_\beta$ satisfies (4), so by assumption, is separated. This implies the two given dotted arrows are the same. Therefore, we have reduced to $X \to S$ is finite type.

Assume $X \to S$ of finite type and assume (4). Since $X \to S$ is finite type, and S is an affine Noetherian scheme, X is also Noetherian (see Morphisms, Lemma 01T6). Therefore, $X \to X \times_S X$ will be a quasi-compact immersion of Noetherian schemes. We proceed by contradiction. Assume that $X \to X \times_S X$ is not closed. Then, there is some $y \in X \times_S X$ in the closure of the image that is not in the image. As X is Noetherian it has finitely many irreducible components. Therefore, y is in the closure of the image of one of the irreducible components $X_0 \subset X$. Give X_0 the reduced induced structure. The composition $X_0 \to X \to X \times_S X$ factors through the closed subscheme $X_0 \times_S X_0 \subset X \times_S X$. Denote the closure of $\Delta(X_0)$ in $X_0 \times_S X_0$ by $\bar{X}_0$ (again as a reduced closed subscheme). Thus $y \in \bar{X}_0$. Since $X_0 \to X_0 \times_S X_0$ is an immersion, the image of X_0 will be open in $\bar{X}_0$. Hence X_0 and $\bar{X}_0$ are birational. Since $\bar{X}_0$ is a closed subscheme of a Noetherian scheme, it is Noetherian. Thus, the local ring $\mathcal{O}_{\bar{X}_0, y}$ is a local Noetherian domain with fraction field K equal to the function field of X_0. By the Krull-Akizuki theorem (see Algebra, Lemma 00PH), there exists a discrete valuation ring A dominating $\mathcal{O}_{\bar{X}_0, y}$ with fraction field K. This allows to construct a diagram:

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X_0 \\ \downarrow & \nearrow & \downarrow \Delta \\ \mathrm{Spec}(A) & \longrightarrow & X_0 \times_S X_0 \end{array} \tag{0.1.2}$$

which sends $\operatorname{Spec} K$ to the generic point of $\Delta(X_0)$ and the closed point of A to $y \in X_0 \times_S X_0$ (use the material in Schemes, Section 01J5 to construct the arrows). There cannot even exist a set theoretic dotted arrow, since y is not in the image of Δ by our choice of y. By categorical means, the existence of the dotted arrow in the above diagram is equivalent to the uniqueness of the dotted arrow in the following diagram:

(0.1.3)
$$\begin{array}{ccc} \operatorname{Spec}(K) & \longrightarrow & X_0 \\ \downarrow & \nearrow & \downarrow \\ \operatorname{Spec}(A) & \longrightarrow & S \end{array}$$

Therefore, we have non-uniqueness in this latter diagram by the nonexistence in the first. Therefore, X_0 does not satisfy uniqueness for discrete valuation rings, and since X_0 is an irreducible component of X, we have that $X \to S$ does not satisfy (4). Therefore, we have shown (4) implies (1). □